# Ashley Nand

## Data & AI Engineer | Generative AI

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**AI/LLM Engineer** building enterprise spend analytics systems with LLM-powered classification, RAG pipelines, and multi-agent workflows.

## Profile Summary

- **Data & AI/ML Engineer** with 3+ years of experience building **enterprise data platforms** and **AI-driven financial analytics solutions** using **Python, PySpark, and Azure Databricks**.
- Designed and implemented **end-to-end ETL/ELT pipelines** to ingest and process **SAP financial spend data** into a governed **Lakehouse architecture**, enabling **SpendCube analytics**, audit-ready reporting, and scalable data modeling.
- Developed **LLM-powered spend classification systems** with **RAG pipelines**, **vendor normalization**, and **human-in-the-loop workflows** using **LangChain, LlamaIndex, vector databases, and FastAPI** to improve data quality, automation, and decision-making.
- Built **scalable APIs and data integration layers** connecting **SAP, data lake, and analytics applications**, reducing manual processes and enabling faster, reliable financial data refresh cycles.

## Core Competencies

- Generative AI, LLMs & Transformer Architectures
- Retrieval-Augmented Generation (RAG) & Semantic Search
- Multi-Agent Systems, AI Workflows & Prompt Engineering
- Vector Databases, Embeddings & LLM Development
- Fine-Tuning, Model Optimization & Evaluation Techniques
- Azure Databricks, PySpark & Distributed Data Processing
- Data Engineering, ETL/ELT Pipelines & Lakehouse Architecture
- FastAPI, REST APIs & Scalable Backend Systems
- Cloud Computing, Docker & Production AI Deployment
- Data Integration, API Orchestration & Scalable Pipelines

## Skills

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|---|---|
| **Languages** | Python, Java, Kotlin, C++, C, TypeScript, JavaScript, SQL |
| **Data Engineering** | Azure Databricks, PySpark, Azure Data Factory (ADF), ETL/ELT Pipelines, Data Modeling, Lakehouse Architecture, Data Governance |
| **Machine Learning & AI** | Scikit-learn, NumPy, Pandas, PyTorch; Generative AI: LLMs, Transformer Architecture |
| **LLM & RAG Systems** | RAG, LangChain, LlamaIndex, Semantic Search, Vector Databases |
| **AI Agents & Deployment** | Multi-Agent Systems, AutoGen, CrewAI, PydanticAI, Model Context Protocol (MCP), Ollama, vLLM, OpenAI API, Gemini API, Groq API |
| **Backend Development** | Node.js, Express.js, FastAPI, RESTful APIs, Spring Boot, Spring MVC |
| **Databases & Analytics** | PostgreSQL, MySQL, MongoDB, Spend Classification, Firebase Realtime DB, SAP Data, Financial Data Processing, Power BI |
| **Cloud & DevOps** | AWS EC2, Docker, CI/CD, Ubuntu Server |
| **Tools** | Git, GitHub, Postman, Android Studio, IntelliJ, PyCharm, VS Code, Cursor AI |

## Technical Experience

### Mindsprint (Olam Group) — Sep 2023 – Present

#### Biocon Biopharmaceutical Client | LLM & Data Engineer

*Bain & Company* | ***Project:*** *AI Spend Classifier / SpendCube (Financial Spend Intelligence)* — Chennai, India

**Responsibilities:**

- Delivered a **financial spend intelligence solution** for **Biocon** by engineering a **production-grade data pipeline** to ingest **SAP direct and indirect spend data** into a governed **enterprise data lake**.
- Designed **scalable data models** and **ETL/ELT workflows** using **Azure Data Factory (ADF)** to support **3-year rolling spend analytics** and **audit-ready reporting**.
- Collaborated with **business and procurement stakeholders** to translate **BRD requirements** into scalable **data architecture and technical solutions** aligned with enterprise standards.
- Conducted **exploratory data analysis (EDA)** on transaction and vendor datasets to identify **data quality issues**, outliers, and mapping gaps, and implemented **data validation frameworks**.
- Developed an **LLM-based classification system using Retrieval-Augmented Generation (RAG) pipelines** to map spend transactions into **standardized L1–L4 taxonomy**, improving **data consistency** and reducing manual classification effort.
- Built **vendor normalization** and **deduplication pipelines** using **LLM and rule-based techniques**, and implemented **human-in-the-loop workflows** with **confidence scoring** to enhance **data quality and governance**.
- Developed and integrated **RESTful APIs** connecting **SAP systems, enterprise data lake, and analytics platforms**, improving **data integration**, reducing manual effort, and accelerating reporting cycles.

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### OFI: Olam Food Ingredients Client | Software Engineer

*Mindsprint (Olam Group)* | ***Project:*** *OFIS Enumerator (**Published on Play Store**)* Chennai, India

**Responsibilities:**

- Led development and deployment of AI and Android projects, combining backend and mobile solutions for enterprise-scale applications.
- Developed the OFIS Enumerator Android app for 10+ million farmers using Java, Kotlin, XML, Jetpack, Room, SQLite, and Spring Boot APIs for secure, real-time data synchronization.
- Built offline-first architecture, improving reliability in low-connectivity regions and reducing crash rates by 40%.
- Enhanced app performance and delivery using Firebase notifications, crash analytics, and optimized CI/CD pipelines with Git, Jenkins, and Gradle.

### TRAINEE | Android Development

*Feb 2023 — Sep 2023*

*Olam Information Services Private Limited* | ***Project:*** *OFIS Enumerator (**Published on Play Store**)* Chennai, India

**Responsibilities:**

- Joined as a trainee and independently managed multiple Android projects, taking full ownership of development, testing, and deployment across the product lifecycle.
- Developed and maintained apps using Java, XML, Room DB, and SQLite, integrating RESTful APIs for seamless backend communication.
- Successfully published the OFIS Enumerator app on Google Play Store, ensuring production readiness, version control with Git, and reliable end-to-end delivery.

## Projects

### Kofi: Study Assistant using CrewAI, RAG, Docker, EC2

*Oct 2025 – Nov 2025*

**Technologies:** Python, CrewAI, LlamaIndex, ChromaDB, Streamlit, FastAPI, Groq, Docker, AWS EC2

Engineered a RAG-based PDF question-answering system leveraging LlamaIndex and ChromaDB for highly efficient vector search. Developed a robust FastAPI + CrewAI backend and an intuitive Streamlit frontend, containerized with Docker and deployed seamlessly on AWS EC2. Optimized embeddings and prompt strategies to deliver precise, context-aware, and reliable responses.

### The Bible Stories

*Nov 2024 – Dec 2024*

**Technologies:** MySQL, React, Express, Node.js, Tailwind CSS, Ubuntu, GitHub

Built a secure, scalable full-stack web application for content management and publication of Bible-related resources. Implemented user authentication, optimized backend APIs, and orchestrated CI/CD pipelines for high-performance deployment.

### Time Series Data Analysis – The University of Auckland, New Zealand

*Nov 2021 – May 2023*

**Technologies:** Machine Learning, Python, Big Data

Conducted advanced time-series modeling, forecasting, and evaluation on real-world datasets. Generated actionable insights for time-dependent data under the mentorship of Dr. Andreas W. Kempa-Liehr.

### Psychological Emotion-Based Classification of EEG Variations (Data Analytics)

*Oct 2021 – Nov 2022*

**Technologies:** Neuroscience, EEG Signals, MATLAB, Big Data

Designed algorithms to classify emotional states from EEG signals by identifying optimal threshold patterns. Extracted and analyzed patterns for accurate emotion recognition across valence, arousal, and dominance dimensions.

### Product Vending Machine with Machine Learning (IoT Project)

*Mar 2022 – May 2022*

**Technologies:** Arduino, Sensor Technology, IoT, Machine Learning

Developed a smart vending machine integrated with IoT sensors and RFID-based authentication. Applied machine learning to analyze seasonal sales trends and generate recommendations for top-selling products.

### Medicare for e-Health Data Management Technologies (Android App)

*Feb 2021 – Jun 2021*

**Technologies:** Android Development, Room DB, Firebase, MVVM

Developed an Android application for patient treatment record management with offline-first storage (Room DB) & real-time cloud sync (Firebase). Ensured maintainable, scalable architecture using MVVM, improving reliability and user accessibility.

## Publications

### Bi-Directional LSTM Classification of Emotions in EEG Signals

**Published: Jan 16, 2023 - IEEE Xplore**

*2022 International Conference on Augmented Intelligence and Sustainable Systems (ICAISS)*, IEEE Xplore. Achieved 94.12% accuracy using Bi-LSTM on DEAP EEG dataset to classify emotions (valence, arousal, dominance) from a single-channel electrode.

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## Certifications

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|---|---|
| NPTEL Online Certification | *Big Data Computing* |
| ICIWRM 2022 | *Int. Conf. on Integrated Water Resources Management* |
| Udemy | *The Data Science Course 2022: Complete Data Science Bootcamp* |
| PSG NI Virtual Instrumentation Center | *Workshop on Real-Time Applications using LabVIEW with Arduino and Raspberry Pi* |
| edX | *Building Modern Node.js Applications on AWS* |
| DevOps Course | *CI/CD (Continuous Integration/Delivery) for Beginners* |
| Microsoft Certification | *Azure Fundamentals (AZ-900)* |
| Udemy | *Learn to Build Microservices using Spring Boot 3, Spring Cloud, React, Kafka, RabbitMQ, REST API, Docker & IntelliJ IDEA* |

## Education

**M.Tech in Artificial Intelligence & Data Science** (CGPA: 8.9) *April 2023*

*Karunya Institute of Technology & Science*

**B.Tech in Computer Science & Engineering** (CGPA: 7.9) *August 2021*

*Jawaharlal Nehru Technological University, Kakinada*

**Higher Secondary Education (CBSE) – PCM** (CGPA: 7.2) *May 2017*

*Kendriya Vidyalaya, Bacheli*